

# W07 - System Testing and UAT

## ScamGuard System/UAT Scenarios Package

Team: ScamGuard - แอปตรวจสอบรูปภาพตัดต่อที่ถูกลำมาหลอกลวง

Course: ENGSE601 - Week 07 Output Candidate

SRS Baseline: v1.1 (2026-09-12)

Code Baseline: Panuwat-ta/project main @ 66bc9e4a

Scope: 12 System Test scenarios + 6 UAT scenarios

Execution: 12 Not Executed + 6 Not Ready; 0 Pass/Fail claimed

Artifact Status: Ready for Review

Date: 2026-09-22

Design and traceability are author-complete. Official Week 07 template confirmation, independent review, deployed environment, System Test execution and 100 real UAT participants are still pending.

# 1. Baseline, Scope and Critical Journeys

| Field                  | Team record                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Requirement authority  | Canonical SRS v1.1, Document/srs, SHA-256 99bb8d...40a                                                                                  |
| Requirement URL        | <a href="https://github.com/Panuwat-ta/project/tree/main/Document/srs">https://github.com/Panuwat-ta/project/tree/main/Document/srs</a> |
| Frozen source baseline | <a href="https://github.com/Panuwat-ta/project">https://github.com/Panuwat-ta/project</a> , main, 66bc9e4a                              |
| Source evidence        | Week 05-06 frozen snapshots, probes and raw outputs                                                                                     |
| Primary Author         | ภาณุวัฒน์                                                                                                                               |
| Assigned Peer Reviewer | เอกพันธ์ - decision/sign-off pending                                                                                                    |
| Course artifact        | System/UAT Scenarios; official template and naming pending                                                                              |

## Critical journeys

| ID  | User goal                  | Trace                         | Main risk                             | Handoff                      |
|-----|----------------------------|-------------------------------|---------------------------------------|------------------------------|
| J01 | First use and access       | FR-AUTH-01-04; FR-PDPA-01     | consent/token behavior                | W05-F03/F04/F05              |
| J02 | Submit image               | FR-SCAN-01/02                 | validation/HTTP contract              | W06-F01                      |
| J03 | Receive trustworthy result | FR-SCAN-03; FR-ANALYSIS-01-04 | incomplete evidence shown as complete | W05-F01/F02; W06-F03/F05/F06 |
| J04 | Understand heatmap         | FR-XAI-01; NFR-06             | missing or misunderstood heatmap      | W06-F04/F07                  |
| J05 | Manage evidence            | FR-HISTORY-01/02              | history/report contract gaps          | W06-F08/F09                  |
| J06 | Exercise privacy rights    | FR-PDPA-01                    | consent/log/deletion gaps             | W05-F04/F05                  |

## Entry criteria for real execution

- approved mobile build id and API deployment URL
- isolated DB, Redis, media storage and model dependencies
- two user accounts, admin context and consented test-data manifest
- source-verification disposition and Week 05-06 defect decisions
- privacy notice, facilitator script and 100 real UAT participants

The local ENGSE212 working tree is develop and was read only. It is not represented as the main baseline; Week 07 reuses the already frozen main evidence.

## 2. System Test Scenarios (1-4 of 12)

| ID    | Trace                            | Scenario                                      | Acceptance oracle                                                   | Status       |
|-------|----------------------------------|-----------------------------------------------|---------------------------------------------------------------------|--------------|
| ST-01 | FR-AUTH-01 AC-1/5;<br>FR-PDPA-01 | Register with system consent false then true  | false rejected; true creates user and consent without password/hash | Not Executed |
| ST-02 | FR-AUTH-02-04; NFR-04            | Login, protected access, refresh and logout   | valid access, 15-minute TTL and invalid/expired rejection match SRS | Not Executed |
| ST-03 | FR-SCAN-01                       | Select image, crop and repeat with skip       | preview matches selected cropped/original image                     | Not Executed |
| ST-04 | FR-SCAN-02 AC-1                  | Upload valid image and poll to terminal state | HTTP 202 + scan id; id/status/progress and result available         | Not Executed |

### Evidence required

- ST-01: sanitized HTTP/DB/consent records
- ST-02: token claims and screen capture
- ST-03: device recording and image checksums
- ST-04: HTTP transcript, timestamps and scan record

Known frozen-baseline risk: system consent was not enforced and scan upload returned HTTP 200 instead of SRS 202. Actual execution must preserve the observed status, not normalize it.

### Execution state

All four scenarios on this page are designed but have not run on an approved integrated build. Pass/Fail count remains zero.

## 2. System Test Scenarios (5-8 of 12)

| ID    | Trace               | Scenario                                             | Acceptance oracle                                                     | Status       |
|-------|---------------------|------------------------------------------------------|-----------------------------------------------------------------------|--------------|
| ST-05 | FR-SCAN-02 AC-2/3/4 | Try unsupported, oversize and over-resolution inputs | each is rejected before analysis with aligned message/status          | Not Executed |
| ST-06 | FR-SCAN-03; NFR-01  | Repeat identical scan with Redis enabled             | miss analyzes; hit reuses usable result/heatmap in <=3 s; TTL 30 days | Not Executed |
| ST-07 | FR-ANALYSIS-01-04   | Run OCR, visual, source and risk pipeline            | ground-truth outputs and Hybrid max+bonus grade match SRS             | Not Executed |
| ST-08 | FR-XAI-01           | Open heatmap; toggle, opacity, zoom and pan          | aligned media and controls/legend/breakdown behave correctly          | Not Executed |

### Evidence required

- ST-05: input manifest/checksums and responses
- ST-06: Redis TTL, media reference and timing log
- ST-07: dataset/model/build manifest and raw JSON
- ST-08: device recording and media checksums

Known frozen-baseline risk: cache/heatmap lifecycle, separate forgery/AI-gen signals and production source verification were incomplete.

### Execution state

All four scenarios on this page are designed but have not run on an approved integrated build. Pass/Fail count remains zero.

## 2. System Test Scenarios (9-12 of 12)

| ID    | Trace                 | Scenario                                         | Acceptance oracle                                              | Status       |
|-------|-----------------------|--------------------------------------------------|----------------------------------------------------------------|--------------|
| ST-09 | FR-HISTORY-01         | List, paginate, search/filter and delete history | newest-first contract and owned data/media lifecycle match SRS | Not Executed |
| ST-10 | FR-HISTORY-02         | Submit report, duplicate and unauthorized cases  | 201/pending; 409 duplicate; unauthorized request rejected      | Not Executed |
| ST-11 | FR-PDPA-01            | Manage consent/log/data and delete account       | withdrawal logged and permitted data removed/anonymized        | Not Executed |
| ST-12 | NFR-04; authorization | Cross-user access and controlled failures        | no disclosure; honest failed/degraded state and safe recovery  | Not Executed |

### Evidence required

- ST-09: API/UI plus before/after DB/media evidence
- ST-10: report/admin queue evidence
- ST-11: redacted consent/account audit evidence
- ST-12: two-user HTTP logs and correlation ids

Known frozen-baseline risk: history/report contracts and consent-management routes require full request/response and lifecycle evidence.

### Execution state

All four scenarios on this page are designed but have not run on an approved integrated build. Pass/Fail count remains zero.

### 3. UAT Protocol and Acceptance Metrics

| Metric                       | Trace       | Threshold             |
|------------------------------|-------------|-----------------------|
| Overall satisfaction         | NFR-06 AC-1 | mean $\geq$ 4.00/5.00 |
| Heatmap meaning Q1           | NFR-06 AC-2 | correct $\geq$ 80%    |
| Red vs green Q2              | NFR-06 AC-2 | correct $\geq$ 80%    |
| Heatmap usefulness Q3        | NFR-06 AC-2 | mean $\geq$ 4.00/5.00 |
| Understand without expert Q4 | NFR-06 AC-2 | Yes $\geq$ 80%        |
| Required sample              | NFR-06      | 100 real participants |

#### Participant and privacy protocol

- target general Android users representing ST01; exclude project-team responses or report separately
- use approved synthetic/licensed images; do not collect real financial slips, faces or personal images
- assign anonymous participant IDs and keep names/emails/raw images out of this repository
- obtain explicit consent before screen/audio recording
- do not explain heatmap answers before Q1-Q4 are completed
- record excluded, abandoned and accessibility-affected sessions transparently

#### Analysis rules

- use valid completed responses as denominator and report exclusions separately
- do not replace missing answers with neutral values
- report counts, denominators, means and distributions
- preserve negative comments; a threshold miss is Fail against that criterion

No participant session has occurred. This package reports no satisfaction score, comprehension percentage or stakeholder acceptance.

## 4. UAT Scenarios and Canonical Questionnaire

| ID     | User task                             | Acceptance observation                                                    | Status    |
|--------|---------------------------------------|---------------------------------------------------------------------------|-----------|
| UAT-01 | Onboarding and register/login         | complete access and identify required system consent without intervention | Not Ready |
| UAT-02 | Select/crop approved image and submit | reach processing/result and understand supported input                    | Not Ready |
| UAT-03 | Explain risk score/evidence           | state screening purpose and non-forensic limitation                       | Not Ready |
| UAT-04 | Use heatmap controls                  | complete toggle/opacity task and Q1-Q4                                    | Not Ready |
| UAT-05 | Find history and report scam          | find prior scan, choose category and see confirmation                     | Not Ready |
| UAT-06 | Satisfaction and safety debrief       | rate experience and identify a safe next action                           | Not Ready |

### Canonical NFR-06 heatmap questions

- 1. บริเวณสีแดงสื่อความหมายถึงอะไร? Correct: จุดที่ระบบประเมินว่าเสี่ยงว่าถูกดัดแปลง
- 2. เปรียบเทียบสีแดงกับสีเขียว ส่วนใดน่าเชื่อถือมากกว่า? Correct: สีเขียว
- 3. Heatmap ช่วยให้คุณมั่นใจในการแยกแยะสลิปปลอมมากขึ้นเพียงใด? Likert 1-5
- 4. ท่านเข้าใจแผนที่ความร้อนโดยไม่ต้องให้ผู้เชี่ยวชาญอธิบายหรือไม่? Yes/No

UAT is Not Ready until an approved build, test-data pack, facilitator script, privacy notice, recruitment record and 100 real participants exist.

## 5. Readiness Evidence and Findings

| Evidence                       | Purpose                                  | State            |
|--------------------------------|------------------------------------------|------------------|
| E07-baseline.txt               | pin SRS hash and frozen code baseline    | Existing         |
| E07-readiness-basis.txt        | carry Week 05-06 frozen source facts     | Existing/derived |
| system_readiness_probe.py      | verify hash, metadata and contract facts | Executed         |
| E07-system-readiness-probe.txt | raw probe output                         | Existing         |
| E07-result-register.txt        | reconcile 18 statuses                    | Existing         |
| System/UAT runtime logs        | prove product/user outcomes              | Not Ready        |
| Human peer-review decision     | independent quality gate                 | Pending          |

### Open findings

| ID  | Finding                                                   | State             | Owner / next action                            |
|-----|-----------------------------------------------------------|-------------------|------------------------------------------------|
| F01 | Official Week 07 template/naming not released in scaffold | Open dependency   | ภานุวัฒน์ - compare and issue v2 if needed     |
| F02 | No approved deployed build/environment id                 | Not Ready         | ภานุวัฒน์ - provision and version all services |
| F03 | Frozen scan/cache/source/heatmap/history/report gaps      | Open product gaps | ภานุวัฒน์ - fix/accept and rerun integration   |
| F04 | Consent/PDPA journey not acceptance-ready                 | Open product gaps | ภานุวัฒน์ - implement and retest               |
| F05 | Controlled test data/ground truth absent                  | Not Ready         | ภานุวัฒน์ - create approved manifest/oracle    |
| F06 | 100 participant sessions do not exist                     | Not Ready         | เอกพันธ์ - recruit, consent and facilitate     |
| F07 | Independent peer-review decision absent                   | Pending           | เอกพันธ์ - review trace/status/PDF/gates       |



## 6. Definition of Done, Review and Handoff

### Author-side design complete

- [x] canonical SRS/version/hash and frozen source baseline pinned
- [x] six critical journeys and eighteen traceable scenarios documented
- [x] twelve System Test scenarios remain Not Executed
- [x] six UAT scenarios remain Not Ready
- [x] readiness findings have owner and next action
- [x] AI use and evidence limitations disclosed
- [x] PDF candidate rendered and checked

### Human/external gates still open

- [ ] official Week 07 template confirmed
- [ ] independent peer-review decision recorded
- [ ] approved system environment and test data available
- [ ] System Test execution evidence and defect disposition recorded
- [ ] 100 real UAT sessions and metric analysis completed
- [ ] merge/tag/course submission event recorded

### Peer review

| Field             | Record                                                    |
|-------------------|-----------------------------------------------------------|
| Reviewer          | เอกพันธ์                                                  |
| Review date       | Pending                                                   |
| Decision          | Pending: Ready / Ready with open issues / Revise          |
| Comments/sign-off | Pending - must be entered after a real independent review |

### AI and evidence declaration

AI assisted organization, trace mapping, readiness analysis, probe/PDF construction and consistency checks. No synthetic Pass/Fail, UAT response, satisfaction score, comprehension percentage, peer approval, stakeholder acceptance, screenshot or runtime log was created. No secret, production data, user image or participant personal data was copied.

Correct current state: Ready for Review. Not Submitted, Accepted or Passed.